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Date: 09/04/2026

Duration: 3 hours

FINAL ASSESSMENT

II YEAR – SEMESTER 4

CSE23AE204 - PCP III

(B.Tech. E01,E02,E03,E05,E06)

SET D

Question 1: Problem Statement

You are developing a **Payroll Processing Module** for a corporate organization.

The system processes payroll details for **one employee at a time**, provided as a **space-separated input string**. Each employee belongs to a specific category and follows predefined salary rules.

The system must:

- Identify the employee type from input
- Compute gross salary using employee-specific rules
- Deduct tax
- Return the final payable salary

Each employee must be assigned a **unique employee ID**, which should be generated internally using a **static variable**.

Salary Calculation Rules

Permanent Employee

$\text{gross} = \text{baseSalary} + \text{hra} - \text{pf}$

$\text{tax} = 10\% \text{ of gross}$

$\text{net} = \text{gross} - \text{tax}$

Contract Employee

$\text{gross} = \text{baseSalary} + \text{allowance}$

$\text{tax} = 10\% \text{ of gross}$

$\text{net} = \text{gross} - \text{tax}$

Example 1

Input

PERMANENT Ram 50000 10000 5000

Output

49500.0

Explanation

$\text{gross} = 50000 + 10000 - 5000 = 55000$

$\text{tax} = 10\% \text{ of } 55000 = 5500$

Question 2: Problem Statement

A company generates **employee badge codes** using characters from a given pool of characters.

For visual symmetry and design standards, the badge code must be a **palindrome**.

You are given a string S containing **uppercase and lowercase English letters**.

Each character can be used **only as many times as it appears** in the string.

You may **rearrange the characters in any order**.

Your Task

Determine the **maximum possible length** of a palindromic badge code that can be generated.

Example 1:

Input

"abccccdd"

Output

7

Explanation:

Palindrome can be "dccaccd"

Question 3: Problem Statement

You are part of an **enterprise risk analytics team**.

A forecasting engine models **hierarchical risk propagation**, where each level derives its values from the **two adjacent values** in the level above it.

To allow automated validation, the engine exports the matrix as:

- A **row-wise hierarchical layout**
- Each row stored as a **string**
- Values computed using previously generated data

Your task is to generate this hierarchical coefficient matrix for a given integer N.

Example 1

Input:

5

Output

```
[ "1",
  "1 1",
  "1 2 1",
  "1 3 3 1",
  "1 4 6 4 1" ]
```

Question 4: Problem Statement

You are given a row of storage units, where each unit contains some amount of resources.

Constraint:

- You **cannot collect from two adjacent units** (security system restriction)

Your task is to determine the **maximum total resources** you can collect without violating the rule.

Example 1

Input: units = [1, 2, 3, 1]

Output: 4

Explanation:

Pick units 1 and 3 $\rightarrow 1 + 3 = 4$

Question 5: Problem Statement

You are given a **directed graph** consisting of n nodes numbered from 0 to n-1.

The graph is represented by a list of **directed edges**, where each edge is represented as:

[u, v]

This means there is a **directed edge from node u to node v**.

Your goal is to find the **minimum set of starting vertices** such that every node in the graph is **reachable from at least one of these starting vertices**.

A node v is considered **reachable** from node u if there exists a path following the directed edges from u to v.

You must return a list of vertices representing the **minimum number of starting nodes** required to reach all nodes in the graph.

The returned list should contain the **minimum number of vertices** required to cover all nodes.

Example 1

Input: n = 6

edges = [[0,1],[0,2],[2,5],[3,4],[4,2]]

Output: [0,3]

Explanation

Graph structure:

0 $\rightarrow$ 1

$\downarrow$

2 $\rightarrow$ 5

3 $\rightarrow$ 4 $\rightarrow$ 2

Nodes **0** and **3** have **no incoming edges**, so they must be included as starting points.

From these two nodes, every other node becomes reachable.